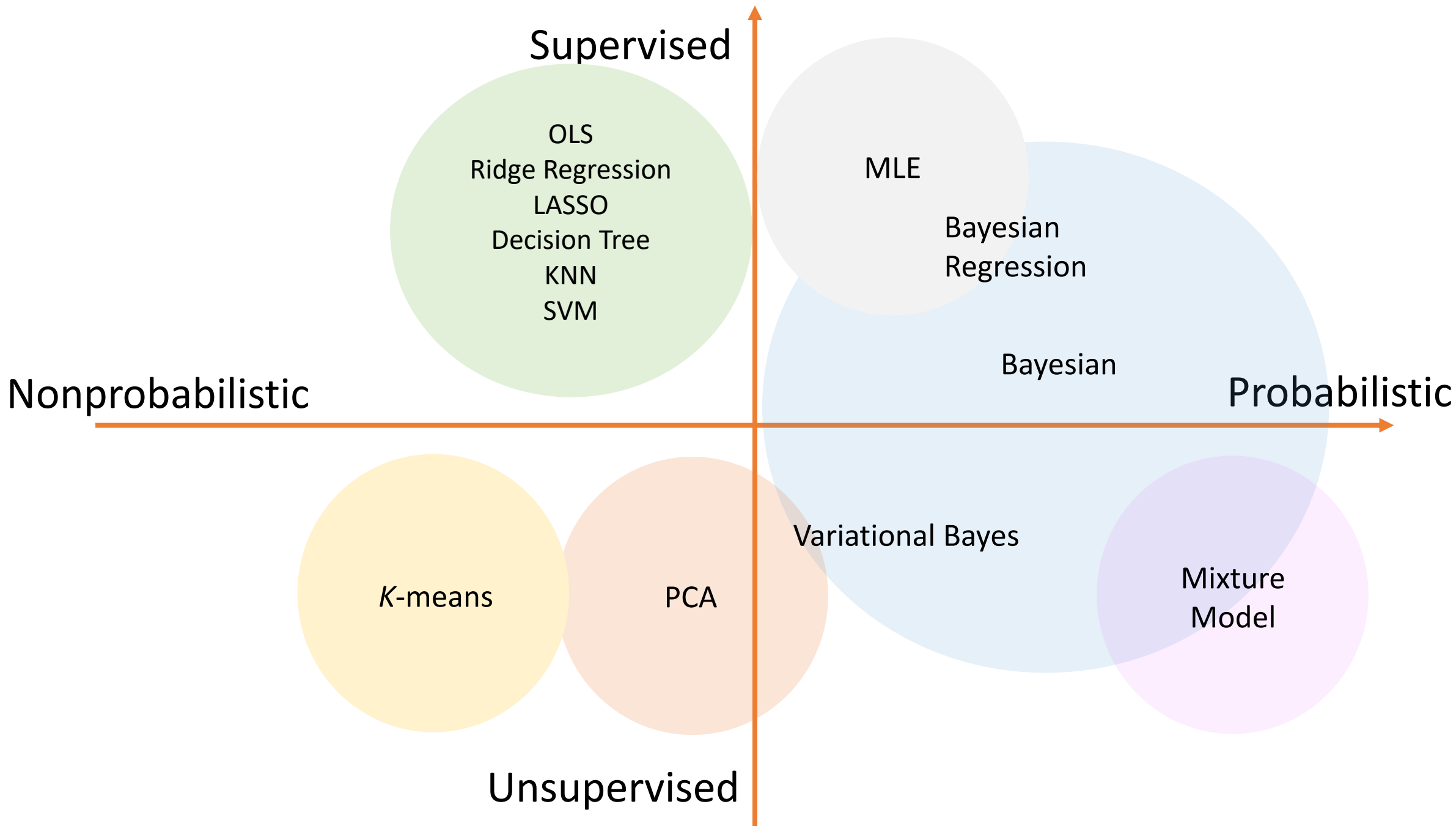


A First Course in Machine Learning

Chapter 5b

Chao-En Yu

National Tsing Hua University



Non-Probabilistic Classifier

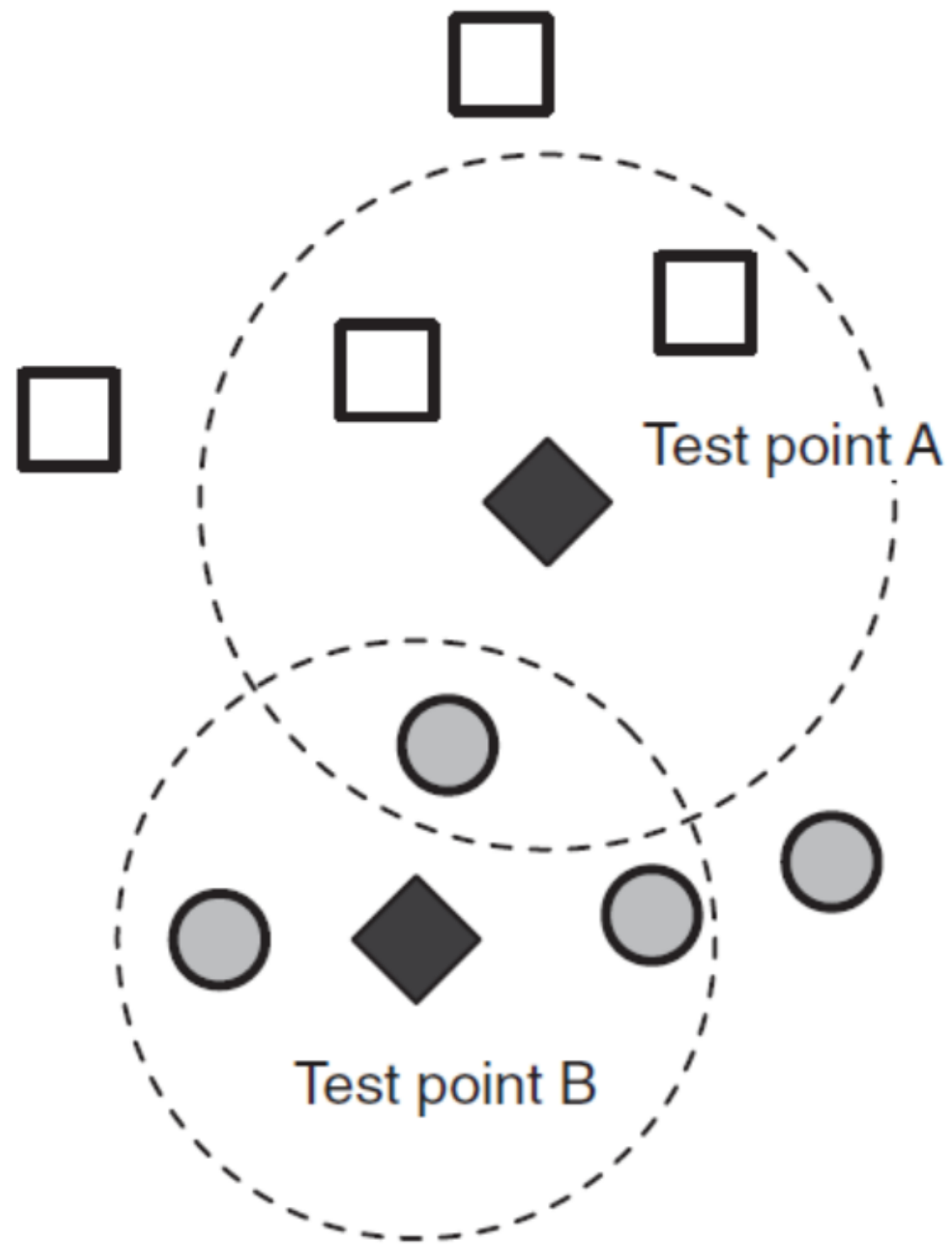
- Rather than providing a probability of class membership, its output is an assignment of an object to a class $t_{\text{new}} = c$.
- Two algorithms are discussed: K -nearest neighbors (KNN) and the Support Vector Machine (SVM).
- The SVM will also provide us with an introduction to the area of kernel methods.

K -Nearest Neighbors (KNN)

- KNN is popular due to simplicity and excellent empirical performance.
- Advantage 1. KNN can handle both binary and multiclass data.
- Advantage 2. KNN makes no assumption about the parametric form of the decision boundary.  Non-parametric
- Advantage 3. KNN does not have a training phase (though we still need training points).

K -Nearest Neighbors (KNN)

- We have N training objects, each of which is represented by a set of attributes $\mathbf{x}_n$ and a label t_n .
- To classify x_{new} with KNN, we first find the K training points that are closest to $\mathbf{x}_{\text{new}}$. t_{new} is then set to be the majority class amongst these neighbors (illustrated in the following figure with $K = 3$).
- The neighbors of test point A include two from the square class and one from the circle class, so it is classified as belonging to the square class.



Issue of Ties

- One drawback of the KNN is *the issue of ties*.
- Not sensible to randomly assign the class from the set of tied classes.
- For binary classification, a solution is to always use an odd number of neighbors.
- Or, weight the votes according to distance such that the votes from closer points have greater influence (Weighted KNN).  Minority may be closer. Can you give an example function?

中獎號碼202

Hi-Life 萊爾富
電子發票證明聯
110年05-06月
NS-05732262

2021-05-09 09:02:42
隨機碼:2828 總計:49
賣方:28414234



門市 3337 序 1-297084 早(0510)
卡號7102109688297580得49點
退貨憑電子發票證明聯正本辦理

Hi-Life 萊爾富
電子發票證明聯
110年05-06月
NS-05733402

2021-05-02 14:38:25
隨機碼:5484 總計:440
賣方:28414234



門市 3337 序 2-468165 早(0503)
卡號7102109688297580得440點
退貨憑電子發票證明聯正本辦理

Hi-Life 萊爾富
電子發票證明聯
110年05-06月
NS-05732022

2021-05-07 11:39:33
隨機碼:3171 總計:56
賣方:28414234



門市 3337 序 1-296700 早(0508)
卡號7102109688297580得56點
退貨憑電子發票證明聯正本辦理

Hi-Life 萊爾富
電子發票證明聯
110年05-06月
NS-05731662

2021-05-06 09:39:47
隨機碼:7234 總計:68
賣方:28414234

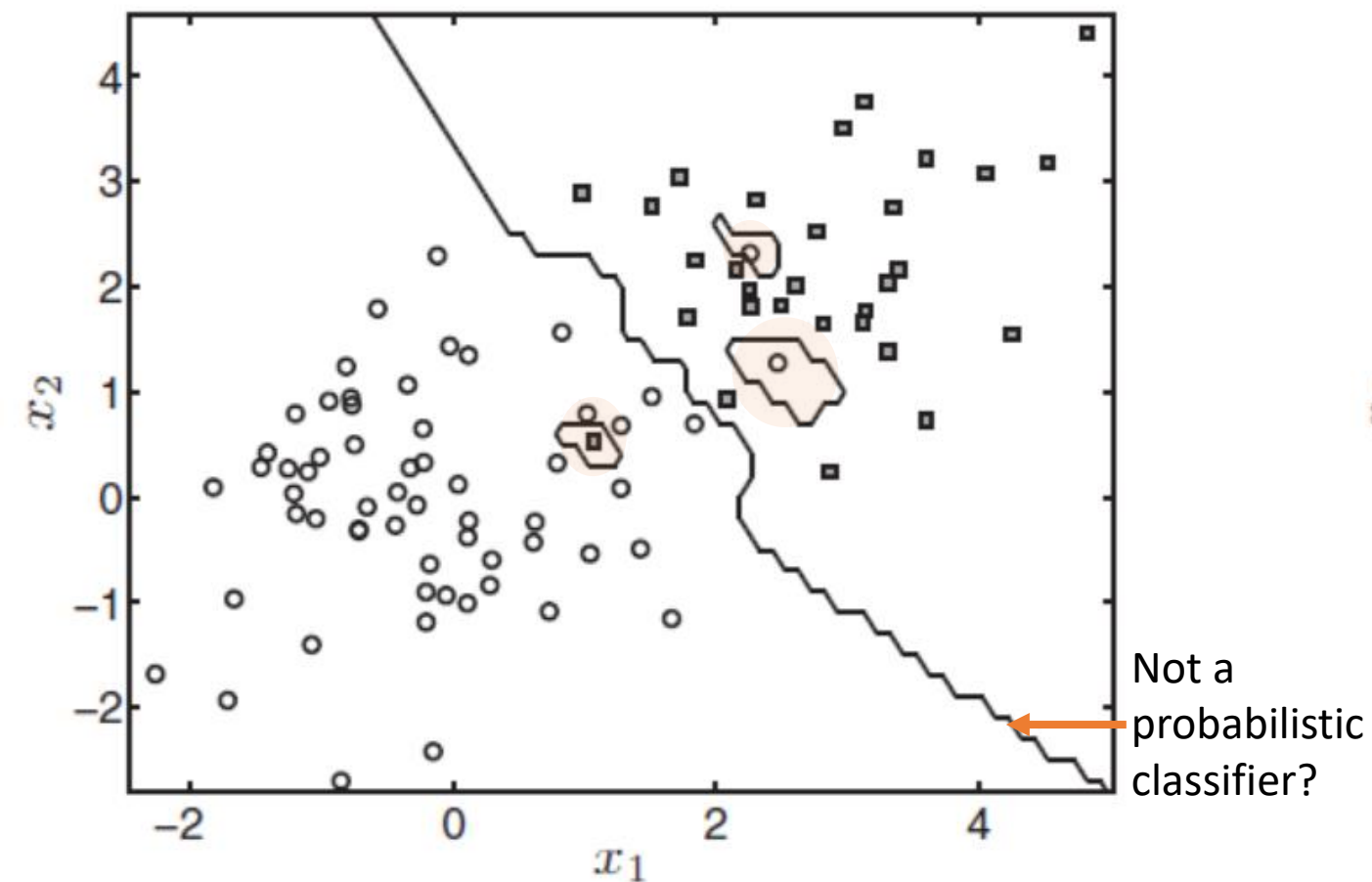


門市 3337 序 1-296085 早(0507)
卡號7102109688297580得68點
退貨憑電子發票證明聯正本辦理

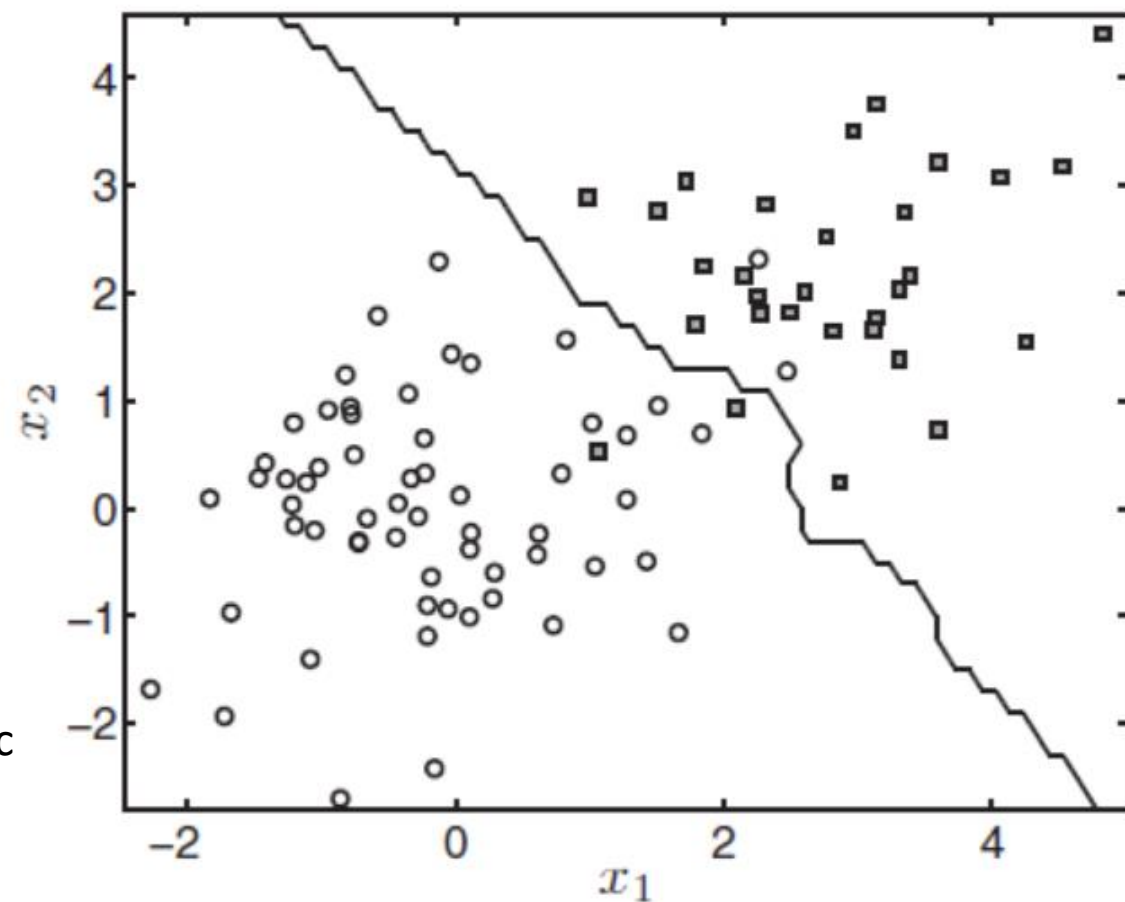
Choosing K

- Once we have some data and have chosen a suitable distance measure, what remains is the choice of K .
- If K is too small, our classification can be heavily influenced by noise (overfitting): When $K = 1$, the three points in the center of the islands are likely to be noise (i.e. mislabeled points).
- The inclusion of more neighbors has had the effect of *regularizing* the boundary, removing the three islands.

Decision Boundaries



(a) Decision boundary when $K = 1$.

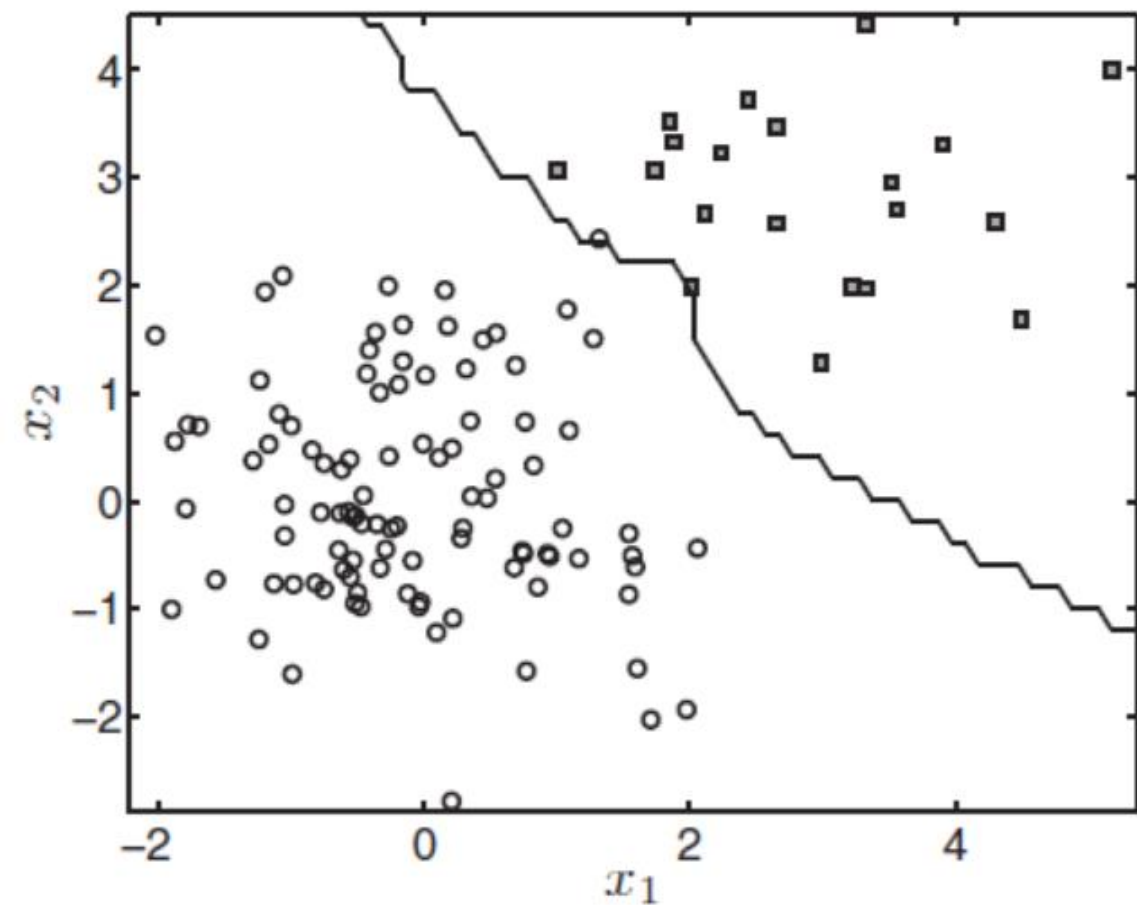


(b) Decision boundary when $K = 5$.

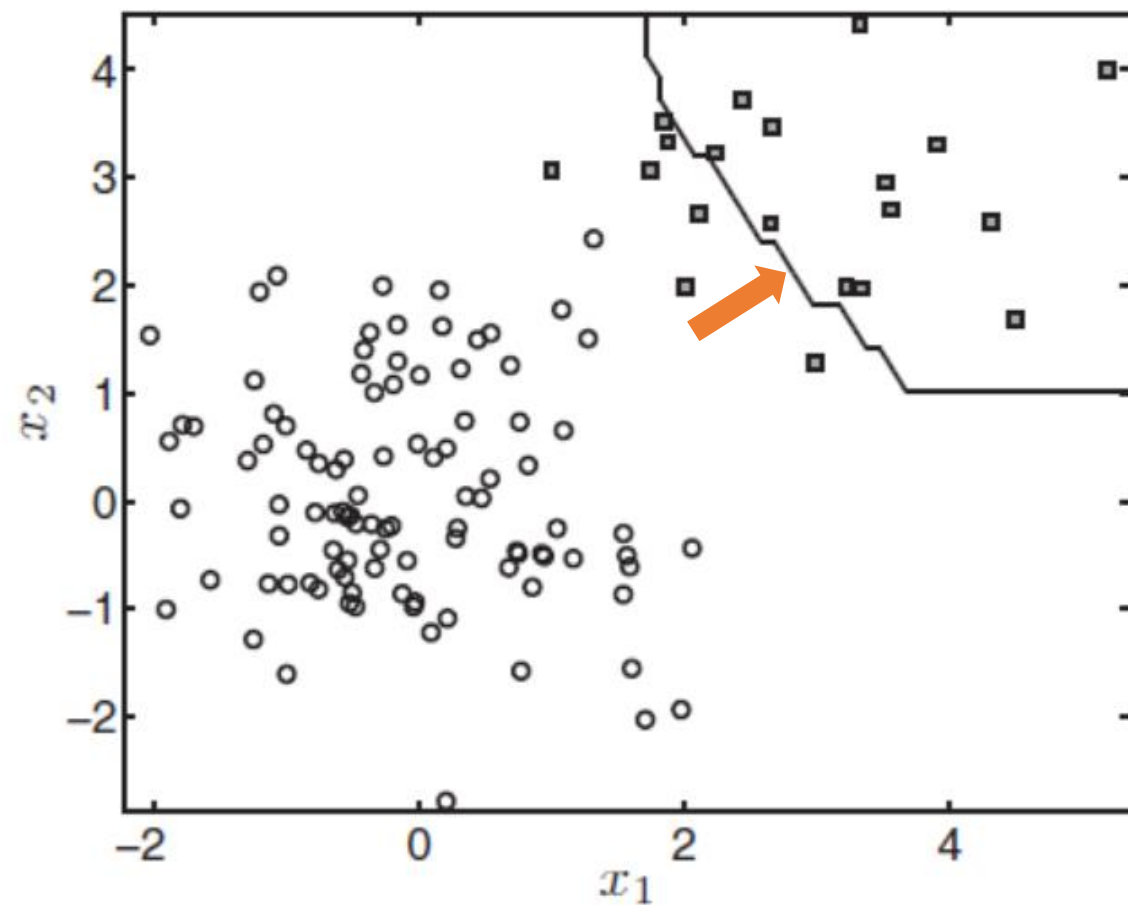
When K is Too Big

- As we increase K , we are using neighbors from further away from $\mathbf{x}_{\text{new}}$.
- Datasets with uneven numbers of objects in each class are known as *imbalanced*.
- When K is too big, we will lose the true patterns in the data:
Either $\mathbf{x}_{\text{new}}$ can never be classified as one specific class, or the boundary is pushed up into the corner (since the larger class exerts too much influence).

Decision Boundaries



(a) Decision boundary when $K = 5$.



(b) Decision boundary when $K = 39$.

Support Vector Machine

- Support Vector Machine (SVM) was introduced by Boser, Guyon, and Vapnik (1992) and Vapnik (1995).
- SVM is particularly useful in applications where the number of attributes is much larger than the number of training objects.
- This is because the number of parameters for the SVM is related to the number of training objects, rather than the number of attributes.

```
newDataset.head()
```

	SCHOOL_YEAR	SCHOOL_CODE	COUNTY	PUBLIC_PRIVATE	CITY	SCHOOL_NAME	REPORTED	ENROLLMENT	CATEGORY	COUNT	PERCENT
0	2016-2017	52749	MONTEREY	PRIVATE	PRUNEDALE	PRUNEDALE CHRISTIAN ACADEMY	Y	10.0	PBE	0.0	NaN
1	2016-2017	52749	MONTEREY	PRIVATE	PRUNEDALE	PRUNEDALE CHRISTIAN ACADEMY	Y	10.0	HEPB	0.0	NaN
2	2016-2017	52749	MONTEREY	PRIVATE	PRUNEDALE	PRUNEDALE CHRISTIAN ACADEMY	Y	10.0	DTP	0.0	NaN
3	2016-2017	52749	MONTEREY	PRIVATE	PRUNEDALE	PRUNEDALE CHRISTIAN ACADEMY	Y	10.0	POLIO	0.0	NaN
4	2016-2017	52749	MONTEREY	PRIVATE	PRUNEDALE	PRUNEDALE CHRISTIAN ACADEMY	Y	10.0	Up-To-Date	0.0	NaN

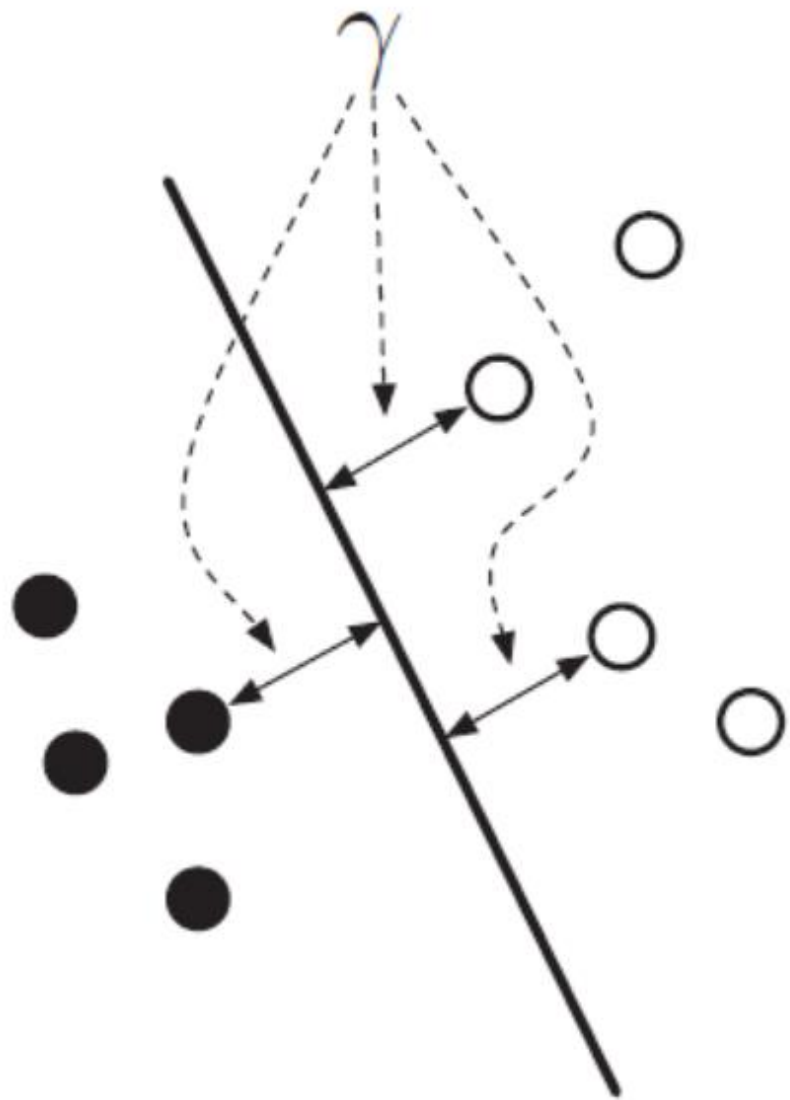
Standard SVM

- The standard SVM uses a linear decision boundary, $\mathbf{w}^T \mathbf{x}_{\text{new}} + b$, to classify new data objects.
- Objects lying on one side of the line are put into class $t_{\text{new}} = 1$ and objects on the other side into $t_{\text{new}} = -1$.
- The SVM decision function for a test point $\mathbf{x}_{\text{new}}$ is therefore given as

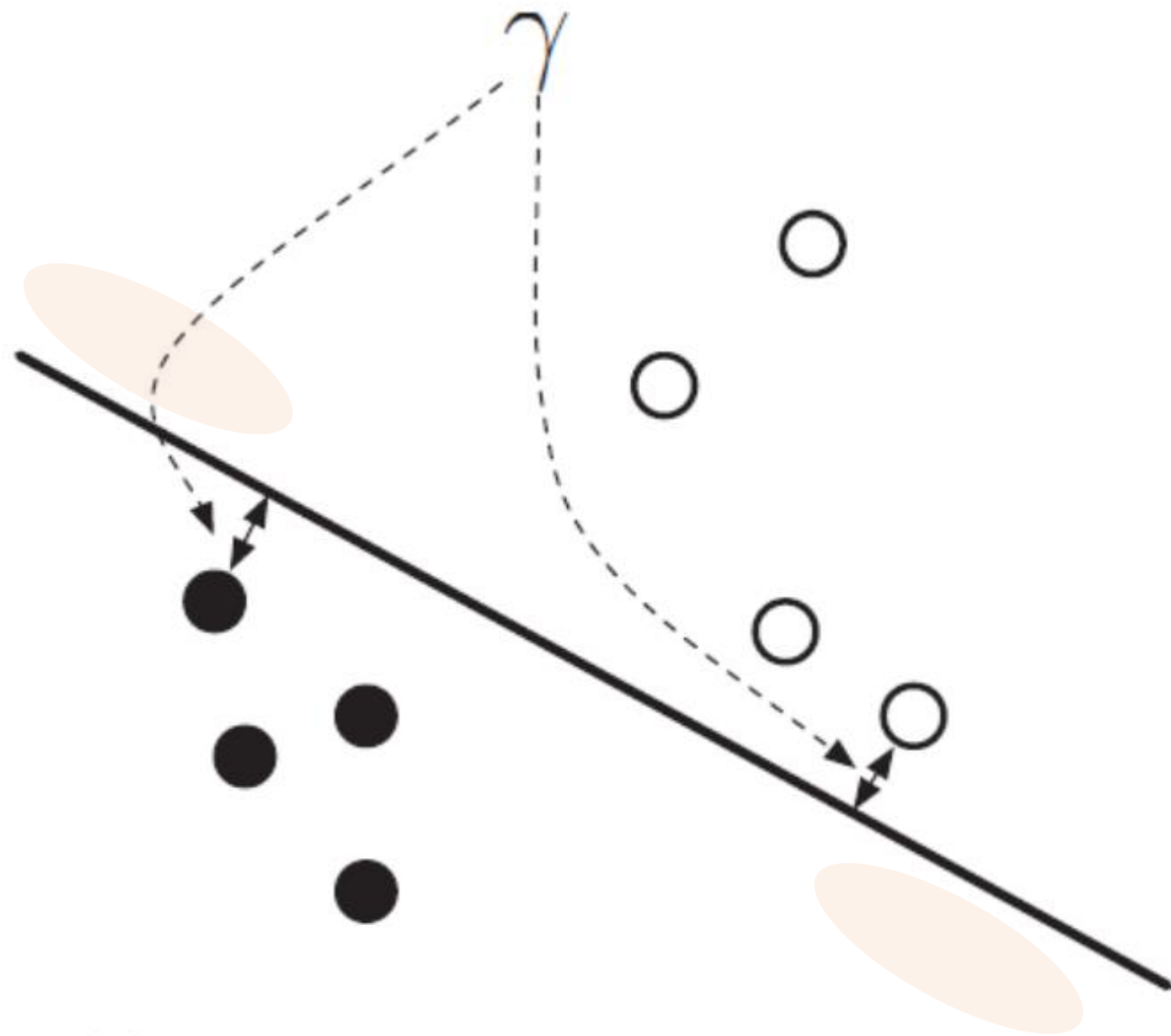
$$t_{\text{new}} = \text{sign}(\mathbf{w}^T \mathbf{x}_{\text{new}} + b)$$

Margin

- The learning task involves choosing the values of $\mathbf{w}$ and b based on training data, which is achieved by maximizing a quantity called *margin*.
 Non-probabilistic
- Margin: The perpendicular distance from the decision boundary to the closest points on either side, denoted by γ .
- Notice that the set of the closed points may *vary* as the boundary changes.



(a) The decision boundary that maximises the margin.



(b) A non-optimal decision boundary.

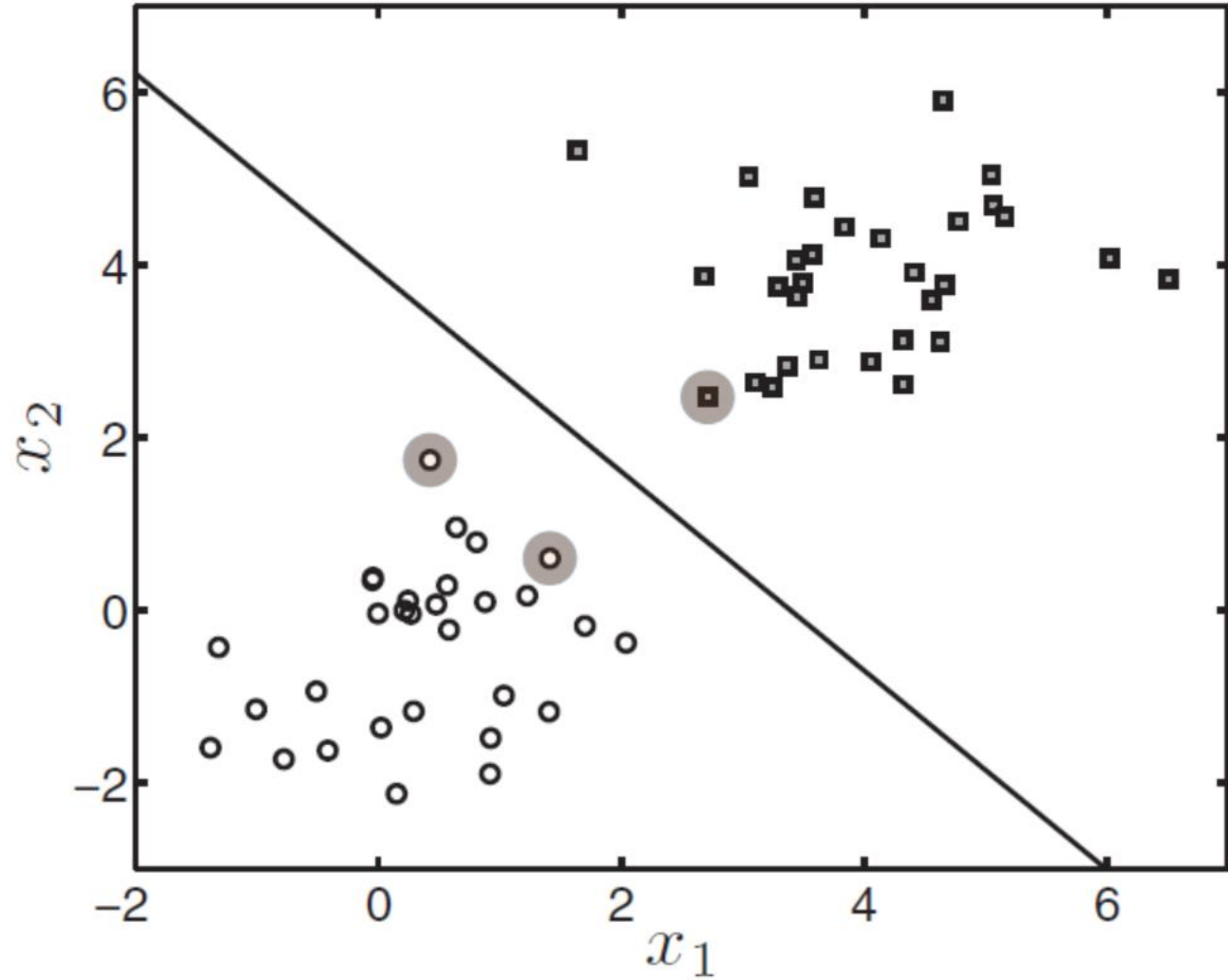
Drawback 1

- As the margin depends only on the closest points, we could discard all of the other data and end up with just the *same* decision boundary.
- At the optimum, all of the α_n that do not correspond to support vectors will be zero.

- This leads to a *sparse solution*—the decision is a function of only a small subset of the training examples (though its calculation is simple).

Especially
comparing with
regression methods

Any idea to solve
this problem?



Drawback 2

- Though it is efficient to use only three of the training points, moving the single data point has had a large effect on the position of the decision boundary.
- This is another example of overfitting—we are allowing the data to have too much influence.
- This happens because of the constraint that all training points *have* to sit on the correct side of the decision boundary, known as a *hard margin SVM*.

